Algebra and Trigonometry **Name (Print):** ANSWER KEY
MATH 1450-02

- This exam contains 12 pages (including the front and back of this cover page) and 14 problems. Check to see if any pages are missing.
- This is the final exam and will count for 20% of your final grade. You have 120 minutes (3:30 pm - 5:30 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 1 bonus questions worth a total of 5 possible points.

Page:	3	4	5	6	7	8	9	10	11	Total
Points:	10	10	10	10	8	8	14	20	10	100
Score:										

Bonus:	

1. Mark each statement as $\boxed{\text{True}}$ or $\boxed{\text{False}}$ by completely shading in the box corresponding to your answer.

   For readability in the key, correct answers are colored in red, e.g. Correct Answer, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

2 pts (a) The conic section represented by the equation $4x^2+16x-16y^2+64y+72=0$ is a parabola.

~~True~~ False

2 pts (b) The conic section represented by the equation $3x^2+5xy+4y^2+1=0$ is an ellipse.

True ~~False~~

2 pts (c) An explicit formula for the arithmetic sequence (starting at a_1) given by $\{-2/3, -1, -4/3, \ldots\}$ is $a_n = -\left(1/3\right)n - 1/3$.

True ~~False~~

2 pts (d) In summation notation, the geometric sum $1+4+16+64+256+1024+4096+16384$ can be expressed as $\sum_{n=0}^{8} 4^n$.

~~True~~ False

2 pts (e) The sequence $\{-1.4, 7, -35, 175, -875, \ldots\}$ is geometric.

True ~~False~~

10 pts 2. Write the equation of the ellipse in standard form, and identify the end points of the major and minor axes as well as the foci.

$$25x^2 - 50x + 16y^2 - 96y - 231 = 0$$

Equation in Standard Form : $\dfrac{(x-1)^2}{4^2} + \dfrac{(y-3)^2}{5^2} = 1$

End Points of Major Axis : $(1,-2)$, $(1,8)$

End Points of Minor Axis : $(-3,3)$, $(5,3)$

Foci : $(1,0)$, $(1,6)$

To begin, we re-write the equation given in the standard form

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1.$$

To do so, we will need to complete the square for both the terms involving x and the terms involving y. This may be accomplished as follows:

$$\begin{aligned}
25x^2 - 50x + 16y^2 - 96y - 231 &= 0\\
25(x^2 - 2x) + 16(y^2 - 6y) - 231 &= 0\\
25(x^2 - 2x + 1 - 1) + 16(y^2 - 6y + 9 - 9) - 231 &= 0\\
25(x^2 - 2x + 1) - 25 + 16(y^2 - 6y + 9) - 144 - 231 &= 0\\
25(x-1)^2 + 16(y-3)^2 - 400 &= 0\\
25(x-1)^2 + 16(y-3)^2 &= 400\\
\frac{25(x-1)^2}{400} + \frac{16(y-3)^2}{400} &= 1\\
\frac{(x-1)^2}{16} + \frac{(y-3)^2}{25} &= 1\\
\frac{(x-1)^2}{4^2} + \frac{(y-3)^2}{5^2} &= 1
\end{aligned}$$

Now, with the standard form of the equation in hand, we can determine the rest of the information requested by the problem. Since $5 > 4$, it follows that the major axis is parallel to the y-axis. So, the coordinates of the vertices (i.e. the end points of the major axis) are $(h, k \pm b)$, i.e. $(1, 3-5) = (1,-2)$ and $(1, 3+5) = (1,8)$. Similarly, the coordinates of the co-vertices (i.e. the end points of the minor axis) are $(h \pm a, k)$, i.e. $(1-4, 3) = (-3,3)$ and $(1+4,3) = (5,3)$. Finally, computing $c = \sqrt{a^2+b^2} = \sqrt{5^2-4^2} = 3$, the coordinates of the foci are $(h, k \pm c)$, i.e. $(1, 3-3) = (1,0)$ and $(1,3+3) = (1,6)$.

10 pts 3. Write the equation of the hyperbola in standard form, and identify the vertices and the foci.

$$16x^2 - 32x - 9y^2 - 36y - 164 = 0$$

Equation in Standard Form : $\dfrac{(x-1)^2}{3^2} - \dfrac{(y+2)^2}{4^2} = 1$

Vertices : $(-2, -2)$, $(4, -2)$

Foci : $(-4, -2)$, $(6, -2)$

To begin, we re-write the equation given in the standard form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

To do so, we will need to complete the square for both the terms involving x and the terms involving y. This may be accomplished as follows:

$$\begin{aligned}
16x^2 - 32x - 9y^2 - 36y - 164 &= 0 \\
16(x^2 - 2x) - 9(y^2 + 4y) - 164 &= 0 \\
16(x^2 - 2x + 1 - 1) - 9(y^2 + 4y + 4 - 4) - 164 &= 0 \\
16(x^2 - 2x + 1) - 16 - 9(y^2 + 4y + 4) + 36 - 164 &= 0 \\
16(x-1)^2 - 9(y+2)^2 - 144 &= 0 \\
16(x-1)^2 - 9(y+2)^2 &= 144 \\
\frac{16(x-1)^2}{144} - \frac{9(y+2)^2}{144} &= 1 \\
\frac{(x-1)^2}{9} - \frac{(y+2)^2}{16} &= 1 \\
\frac{(x-1)^2}{3^2} - \frac{(y+2)^2}{4^2} &= 1
\end{aligned}$$

Now, with the standard form of the equation in hand, we can determine the rest of the information requested by the problem. Since the term involving x's is positive and the term involving the y's is negative, it follows that the transverse axis is parallel to the x-axis. So, the coordinates of the vertices are $(h \pm a, k)$, i.e. $(1-3, -2) = (-2, -2)$ and $(1+3, -2) = (4, -2)$. Finally, computing $c = \sqrt{a^2 + b^2} = \sqrt{3^2 + 4^2} = 5$, the coordinates of the foci are $(h \pm c, k)$, i.e. $(1-5, -2) = (-4, -2)$ and $(1+5, -2) = (6, -2)$.

10 pts 4. Rewrite the given equation in standard form, and then determine the vertex, focus, directrix, and axis of symmetry of the parabola.

$$2y^2 + 8y + 8x - 24 = 0$$

Equation in Standard Form : $(y+2)^2 = 4(-1)(x-4)$

Vertex : $(4, -2)$

Directrix : $x = 5$

Axis of Symmetry : $y = -2$

To begin, we re-write the equation given in the standard form

$$(y-k)^2 = 4p(x-h).$$

To do so, we will need to complete the square for the terms involving y. This may be accomplished as follows:

$$\begin{aligned}
2y^2 + 8y + 8x - 24 &= 0 \\
2(y^2 + 4y) + 8x - 24 &= 0 \\
2(y^2 + 4y + 4 - 4) + 8x - 24 &= 0 \\
2(y^2 + 4y + 4) - 8 + 8x - 24 &= 0 \\
2(y+2)^2 + 8x - 32 &= 0 \\
2(y+2)^2 &= -8x + 32 \\
(y+2)^2 &= -4x + 16 \\
(y+2)^2 &= -4(x-4)
\end{aligned}$$

Now, with the standard form of the equation in hand, we can determine the rest of the information requested by the problem. As always, the coordinate of the vertex is $(h,k) = (4,-2)$. Since the term involving y's is squared and $p = -1 < 0$, it follows that the parabola opens to the left. So, the coordinate of the focus is $(h+p, k) = (4 + (-1), -2) = (3, -2)$ and the directrix is $x = h - p$, i.e. $x = 4 - (-1) = 5$. Finally, the axis of the symmetry is $y = k$, i.e. $y = -2$.

3 pts 5. Write the first six terms of the sequence.

$$a_1 = -2, \;\; a_2 = 5, \;\; a_n = a_{n-2}(4 - a_{n-1})$$

Using the recursive definition of the sequence we are given, we compute

$$\begin{aligned}
a_3 &= a_1(4 - a_2) = (-2)\big(4 - (5)\big) = 2 \\
a_4 &= a_2(4 - a_3) = (5)\big(4 - (2)\big) = 10 \\
a_5 &= a_3(4 - a_4) = (2)\big(4 - (10)\big) = -12 \\
a_6 &= a_4(4 - a_5) = (10)\big(4 - (-12)\big) = 160
\end{aligned}$$

So, the first six terms of the sequence are

$$-2, 5, 2, 10, -12, 160$$

5 pts 6. Write the first five terms of the arithmetic sequence given the following two terms

$$a_{11} = -185, \;\; a_{31} = -485$$

Recall that the recursive formula for an arithmetic sequence is $a_n = a_{n-k} + kd$. Using what we have, this gives

$$\begin{aligned}
a_{31} &= a_{11} + (31 - 11)d \\
-485 &= -185 + 20d \\
-300 &= 20d \\
d &= -15
\end{aligned}$$

So, $a_1 = a_1 1 - 10d = -185 - 10(-15) = -35$. Using the explicit formula for an arithmetic sequence, $a_n = a_1 + (n-1)d$, this gives

$$\begin{aligned}
a_2 &= -50 \quad & a_3 &= -65 \\
a_4 &= -80 \quad & a_5 &= -95
\end{aligned}$$

So, the first five terms of the arithmetic sequence are

$$-35, -50, -65, -80, -95$$

$\boxed{3 \text{ pts}}$ 7. Starting with a_1, write the first five terms of the geometric sequence

$$a_n = -2(3)^{n-1}$$

Using the explicit definition of the geometric sequence we are given, we compute

$$a_1 = -2(3)^{1-1} = -2$$
$$a_2 = -2(3)^{2-1} = -6$$
$$a_3 = -2(3)^{3-1} = -18$$
$$a_4 = -2(3)^{4-1} = -54$$
$$a_5 = -2(3)^{5-1} = -162$$

So, the first six terms of the sequence are$-2, -6, -18, -54, -162$

$\boxed{5 \text{ pts}}$ 8. Find the 11^{th} term in the arithmetic sequence below.

$$\{-10, -13, -16, \dots\}$$

Since this is an arithmetic sequence, we begin by computing the common difference:

$$d = (-13) - (-10) = -3.$$

So, since the explicit formula for an arithmetic sequence is $a_n = a_1 + (n-1)d$, we have

$$a_{11} = -10 + (11-1)(-3) = -10 - 30 = -40.$$

Therefore, the 11^{th} term of the arithmetic sequence we are given is $a_{11} = -40$.

4 pts 9. Write an explicit formula for the geometric sequence

$$\left\{2, 1, \frac{1}{2}, \frac{1}{4}, \ldots\right\}$$

Since this is a geometric sequence, we begin by computing the common ratio:

$$r = \frac{1}{2} = \frac{1/2}{1/4} = \frac{1}{2}.$$

Since the explicit formula for a geometric sequence is of the form $a_n = a_1 r^{n-1}$, an explicit formula for the geometric sequence were are given is

$$a_n = 2\left(\frac{1}{2}\right)^{n-1}.$$

10 pts 10. Find the value of the sum

$$1 + 2 + 3 + \ldots + 999 + 1000.$$

Since $\{1, 2, 3, \ldots, 999, 1000\}$ is an arithmetic sequence, this is a finite arithmetic series of the form $\sum_{n=1}^{1000} a_n$ with $a_n = n$. So, we use the formula for the sum of the first n terms of an arithmetic series, i.e.

$$S_n = \frac{n(a_1 + a_n)}{2}.$$

Since $n = 1000$, a, $a_1 = 1$, and $a_{1000} = 1000$ here, we have

$$S_n = \frac{1000(1 + 1000)}{2} = 500(1001) = 500,500.$$

So, we reach our final answer:

$$1 + 2 + 3 + \ldots + 999 + 1000 = 500,500.$$

10 pts 11. Find the value of the sum

$$-1 - \frac{1}{4} - \frac{1}{16} - \frac{1}{64} - \dots$$

Since $\left\{-1, -\frac{1}{4}, -\frac{1}{16}, \dots\right\}$ is a geometric sequence, this is a geometric series of the form $\sum_{n=1}^{\infty} a_1 r^{n-1}$ with $r = {}^1\!/_4$. So, since $r < 1$, this series converges to a finite value and we may use the formula for the sum of a geometric series, i.e.

$$\sum_{n=1}^{\infty} a_1 r^{n-1} = \frac{a_1}{1-r}.$$

Since $a_1 = -1$ and $r = 14$ here, we have

$$\sum_{n=1}^{\infty} -1 \left(\frac{1}{4}\right)^{n-1} = \frac{-1}{1 - ({}^1\!/_4)} = \frac{-1}{{}^3\!/_4} = -\frac{4}{3}.$$

So, we reach our final answer:

$$-1 - \frac{1}{4} - \frac{1}{16} - \frac{1}{64} - \dots = -\frac{4}{3}.$$

10 pts 12. Find the sum of the finite geometric series

$$\sum_{k=1}^{12} 3\left(\frac{1}{4}\right)^{k-1}$$

Recall that the sum of a finite geometric series is given by

$$\sum_{k=1}^{n} a_1 (r)^{k-1} = \frac{a_1(1-r^n)}{1-r}.$$

So,the sum of the finite geometric series we are given is

$$\sum_{k=1}^{12} 3\left(\frac{1}{4}\right)^{k-1} = \frac{3\left(1 - {}^1\!/_4{}^{12}\right)}{1 - {}^1\!/_4} \approx 4$$

10 pts 13. Find the sum of the infinite geometric series

$$\sum_{k=1}^{\infty} -\frac{9}{4}\left(-\frac{4}{9}\right)^{k-1}$$

Recall that the sum of an infinite geometric series with common ratio r such that $-1 < r < 1$ is given by

$$\sum_{k=1}^{\infty} a_1(r)^{k-1} = \frac{a_1}{1-r}.$$

So, since $-1 < -4/9 < 1$, the sum of the infinite geometric series we are given is

$$\sum_{k=1}^{\infty} -\frac{9}{4}\left(-\frac{4}{9}\right)^{k-1} = \frac{-9/4}{1-(-4/9)} = -\frac{81}{52}$$

5 (bonus) 14. Compute the derivative of $f(x) = x^2 - 4$ using using *only* the limit definition of the derivative,

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}.$$

Using the definition of the derivative, we calculate:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{[(x+h)^2 - 4] - [x^2 - 4]}{h} \\ &= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - 4 - x^2 + 4}{h} \\ &= \lim_{h\to 0} \frac{2xh + h^2}{h} \\ &= \lim_{h\to 0} 2x + h \\ &= 2x. \end{aligned}$$